

PRNet: PAPR Reduction via Deep Autoencoder



The Autoencoder Paradigm Shift

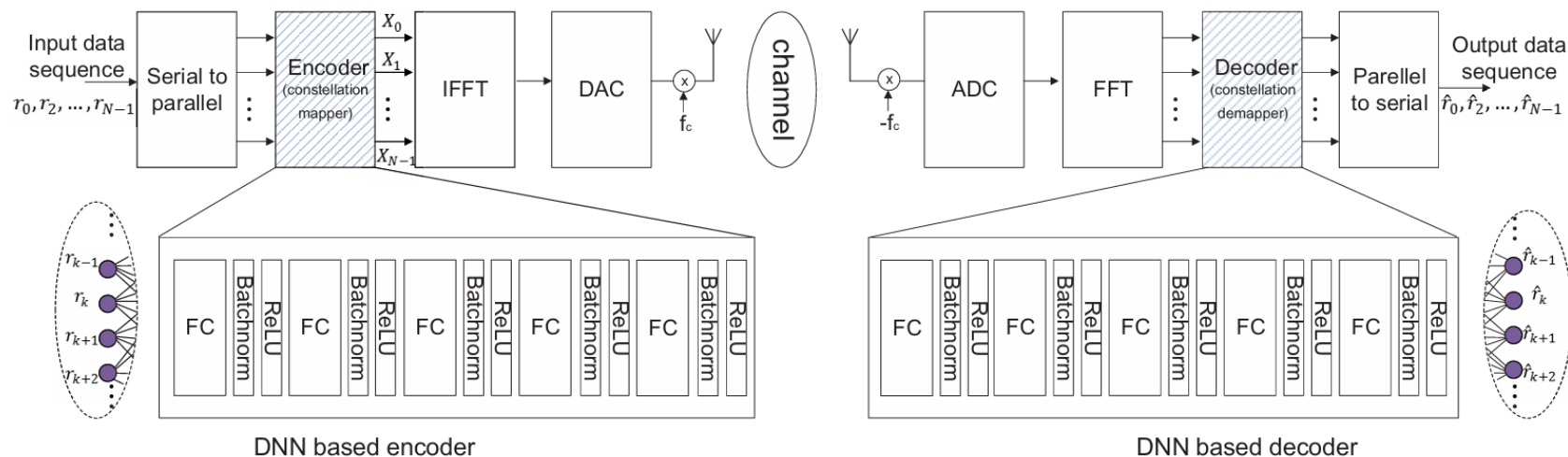
Introduced by Kim et al., PRNet is the first deep learning-based system for joint PAPR and BER optimization. It treats the entire transceiver chain as a single deep autoencoder:

- Encoder → Neural Network Transmitter
- Decoder → Neural Network Receiver
- Bottleneck → Noisy wireless channel

Both the transmitter and receiver are jointly trained, allowing the network to:

- Learn a custom, data-dependent constellation mapping that jointly minimizes BER and PAPR.
- Eliminate side information (SI) overhead entirely — the decoder implicitly learns the encoder's mapping.
- Bypass the exponential search spaces of classical methods (e.g. PTS).

PRNet Transceiver Architecture



End-to-End Neural Network:

- Standard modulation and demodulation are completely replaced by a DNN Encoder and DNN Decoder.
- Structure: Input layer (128 nodes) → 5 hidden layers (2048 nodes each) → Output layer (128 nodes).

Inside each hidden layer block:

- **Fully Connected (FC):** Links all data to process subcarriers jointly.
- **Batch Normalization (BN):** Scales data to ensure stable and reliable training.
- **ReLU Activation:** Introduces the non-linearity required to solve the highly complex trade-off between PAPR and BER

End-to-End System Model

1. **Input Representation: Real I/Q vector**

$$\mathbf{r} = [I_0, Q_0, \dots, I_{N-1}, Q_{N-1}]^T \in \mathbb{R}^{2N}$$

2. **DNN Encoder: parameterized by weights**

$$\mathbf{X} = f(\mathbf{r}; \theta_f) \in \mathbb{C}^N$$

3. **OFDM Mod & Channel: IFFT, then Rayleigh fading & AWGN**

$$y[n] = \mathbb{H}(\text{IFFT}(\mathbf{X}))$$

4. **Demodulation & Equalization: FFT to frequency domain, zero-forcing**

$$\hat{Y}_k = X_k + \frac{E_k}{H_k}$$

5. **DNN Decoder: parameterized by weights**

$$\hat{\mathbf{r}} = g(\hat{\mathbf{Y}}; \theta_g) \in \mathbb{R}^{2N}$$

6. **Fully Differentiable End-to-End Chain**

$$\hat{\mathbf{r}} = \underline{g(\text{FFT}(\mathbb{H} \circ \text{IFFT}(f(\mathbf{r}; \theta_f)) + \epsilon); \theta_g)}$$

The Joint Loss Function

Reconstruction loss (MSE in $\mathbb{R}^{2N}$):

$$\mathcal{L}_1(\mathbf{r}, \hat{\mathbf{r}}) = \frac{1}{2N} \left\| \mathbf{r} - g(\text{FFT}(\mathbb{H} \circ \text{IFFT}(f(\mathbf{r}; \theta_f)) + \epsilon); \theta_g) \right\|_2^2 = \frac{1}{2N} \sum_{i=0}^{2N-1} (r_i - \hat{r}_i)^2.$$

PAPR penalty (oversampled time-domain waveform, $L \geq 4$)

$$\mathcal{L}_2(\mathbf{r}) = \text{PAPR} \{ \text{IFFT}(f(\mathbf{r}; \theta_f)) \} = \frac{\max_{0 \leq n \leq NL-1} |x[n]|^2}{\frac{1}{NL} \sum_{n=0}^{NL-1} |x[n]|^2}$$

Joint loss: (λ governs the BER–PAPR trade-off)

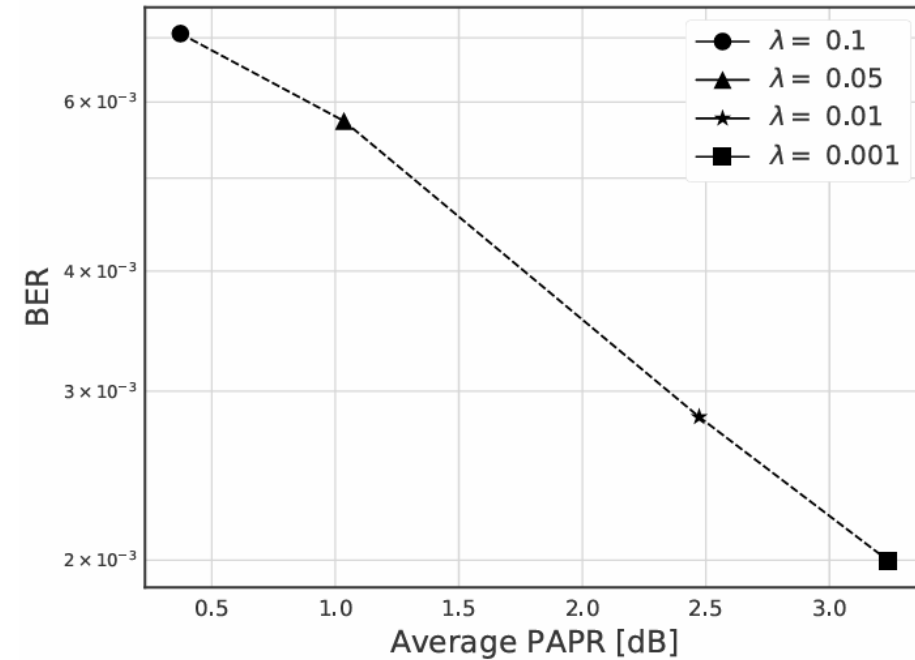
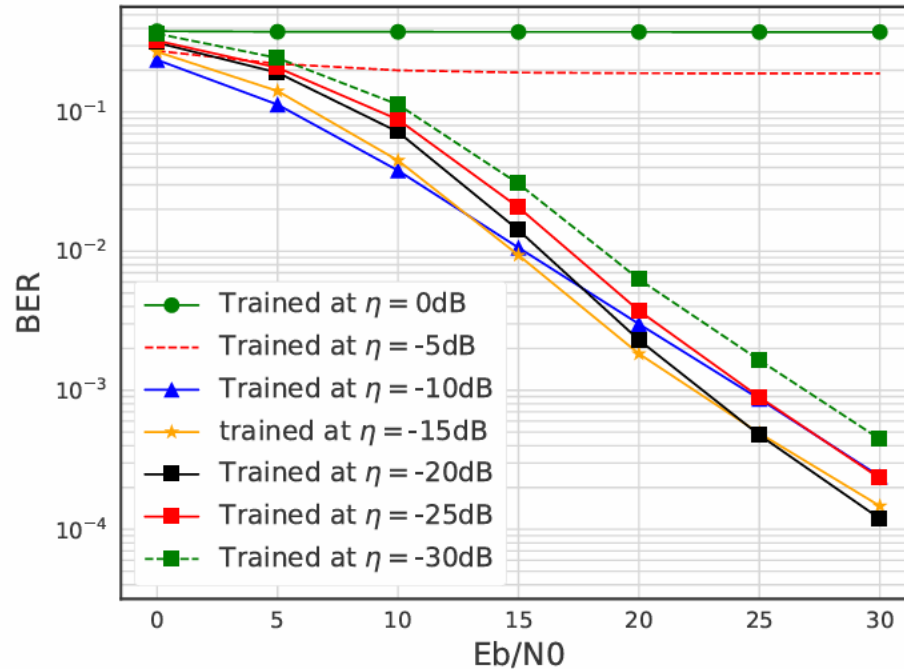
$$\mathcal{L}(\mathbf{r}, \hat{\mathbf{r}}) = \mathcal{L}_1(\mathbf{r}, \hat{\mathbf{r}}) + \lambda \mathcal{L}_2(\mathbf{r})$$

Two-Stage Training Methodology

Denoising autoencoder: Artificial noise at a corruption level η (noise-to-signal ratio) is injected during the forward pass to simulate the physical channel and force the decoder to learn robust decision boundaries.

Stage 1 — Noise Selection ($\lambda = 0$): Sweep η while training with reconstruction loss $\mathcal{L}_1$ only.

Stage 2 — Trade-Off Training ($\eta = \eta^*$): Sweep penalty weight λ under the full loss $\mathcal{L} = \mathcal{L}_1 + \lambda \mathcal{L}_2$



Simulation Setup & Parameters

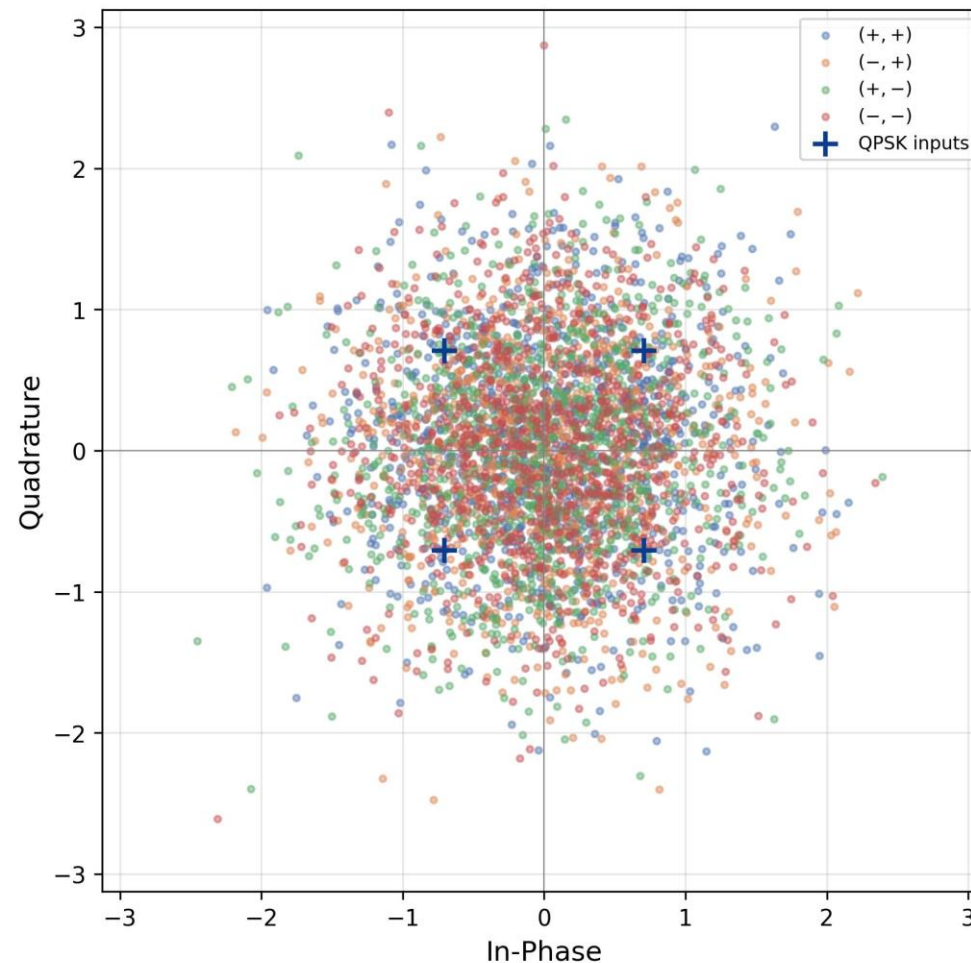
Parameter	Value
Number of Subcarriers (N)	64
Modulation Scheme	QPSK (4-QAM)
DNN Architecture	5 Hidden Layers, 2048 Nodes/Layer
Activation Functions	ReLU (Hidden), Linear (Output)
Optimizer	Adam
Learning Rate	10^{-4}
Batch Size	400
Training Steps	500,000
Optimal Corruption (η^*)	-15 dB
PAPR Penalty Weight (λ)	0.01

Learned Constellation Shaping

Unlike classical OFDM with rigid QPSK boundaries, the PRNet encoder dynamically disperses constellation points:

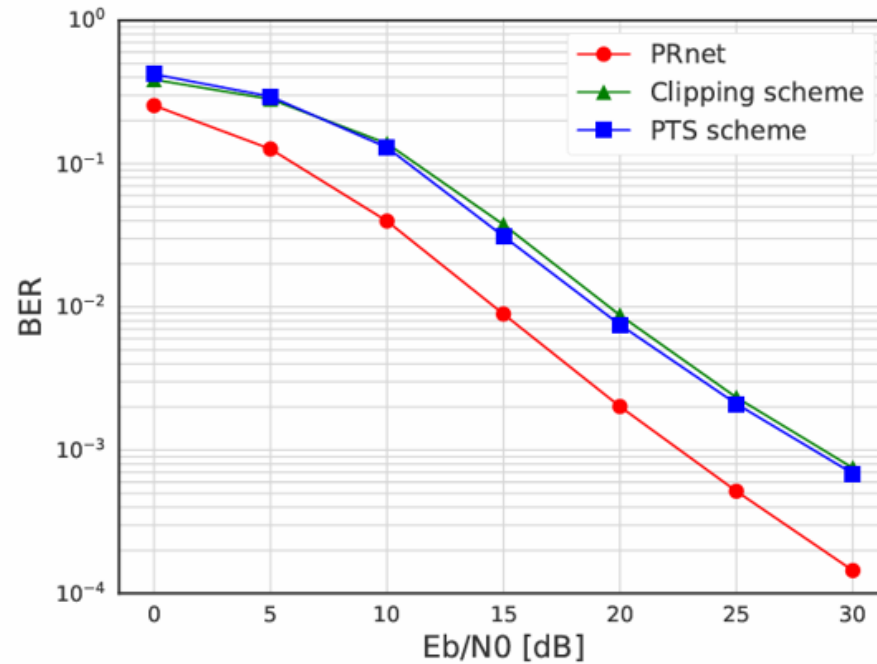
- Maps discrete input data into a continuous, cloud-like spatial distribution.
- Trades traditional Euclidean spacing for optimal PAPR reduction.
- The jointly trained decoder learns to recover data from this custom distribution without explicit knowledge of the mapping.

PRNet Encoder Constellation

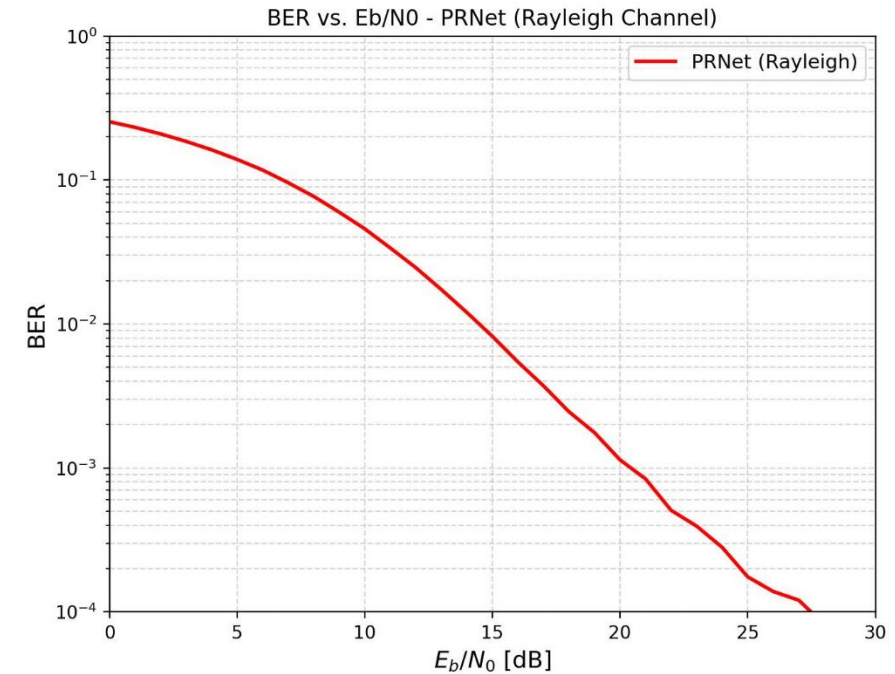


BER Performance

Original Paper



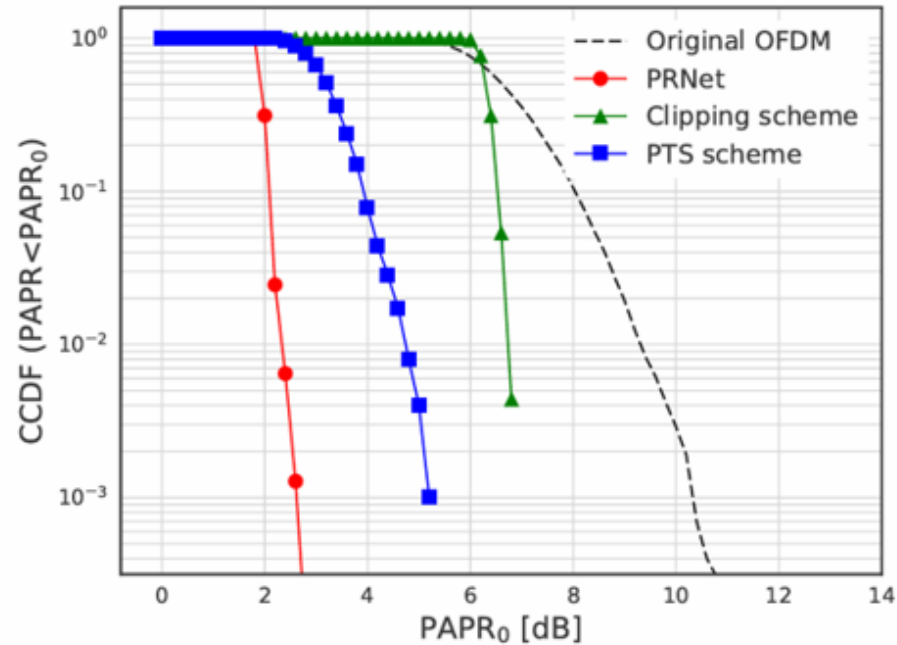
Our Reproduction



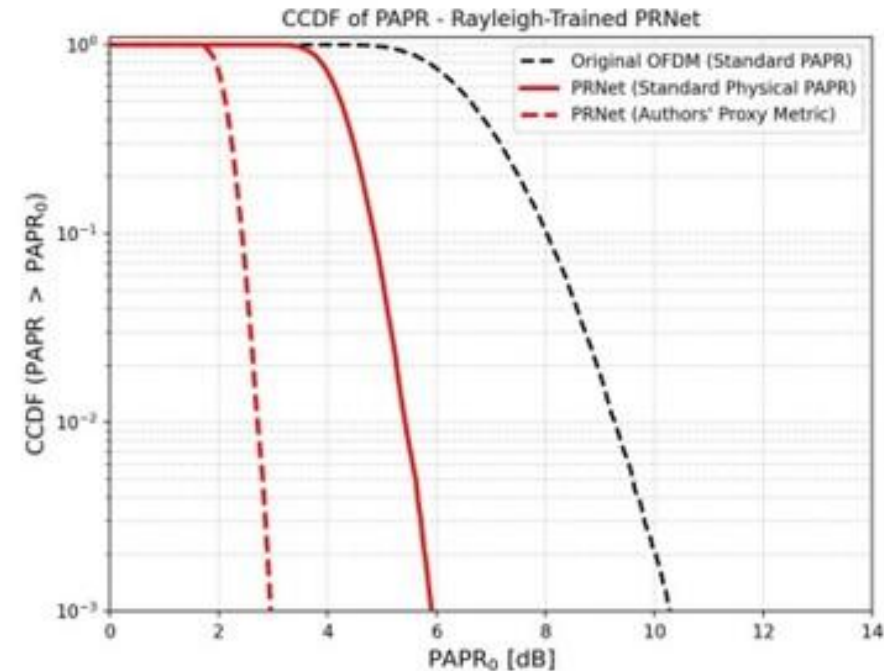
- **Accurate Reproduction:** Our independently trained model perfectly matches the BER curve from the original paper.
- **Superior to Classical Methods:** PRNet achieves a lower Bit Error Rate (BER) than traditional techniques like Clipping and PTS.
- **Proven Robustness:** The results confirm that the neural network successfully learns to protect the data against severe channel fading.

PAPR Performance

Original Paper



Our Reproduction



- **Massive Reduction:** PRNet successfully reduces PAPR by an impressive 4–5 dB compared to standard OFDM (≈ 10.5 dB).
- **A Surprising Discrepancy:** The original paper reports a final PAPR of ≈ 3 dB. However, our strict reproduction reveals the true physical PAPR is actually ≈ 6 dB.

PAPR Reduction – Discrepancy Investigated

To understand why our results exactly doubled the original paper's claims, we audited the authors' open-source implementation.

Root Cause Discovered in the Authors' GitHub Repository

Line 77: `peak_power = tf.reduce_max(tf.abs(encoded_symbol))`

↪ *Extracts peak amplitude instead of physical peak power.*

Line 251: `... z < 10 * np.log10(PAPR_sample) ...`

↪ *Applies $10 \log_{10}$ to an amplitude instead of $20 \log_{10}$. This halves the reported dB.*

- **The Mathematical Impact:** This coding error artificially halves the reported values, making a true 6 dB PAPR look like 3 dB.
- **The Reality:** Even after fixing this mathematical error, PRNet still reduces PAPR by an impressive 4–5 dB compared to standard OFDM.

Computational Complexity & Limitations

Computational Complexity:

- **Theoretical complexity $\mathcal{O}(L \cdot k^2)$.**
- **Real-time GPU inference: 480 μs vs. PTS: 2,890 μs ($\approx 6\times$ speedup over classical methods).**

Practical Limitations of PRNet:Type equation here.

- **Scalability:** FC layers scale as $\mathcal{O}(N^2)$ — impractical for commercial $N = 1024$.
- **Channel mismatch:** Overfits to the training fading model; degrades under unmodelled conditions.
- **Configuration rigidity:** Changing N or M requires full architectural redesign and retraining.
- **Interpretability:** Black-box weights prevent worst-case guarantees required by telecom standards.